SHRI G.S. INSTITUTE OF TECHNOLOGY AND SCIENCE, INDORE - 3

Name : HARSH RAJ SHARMA

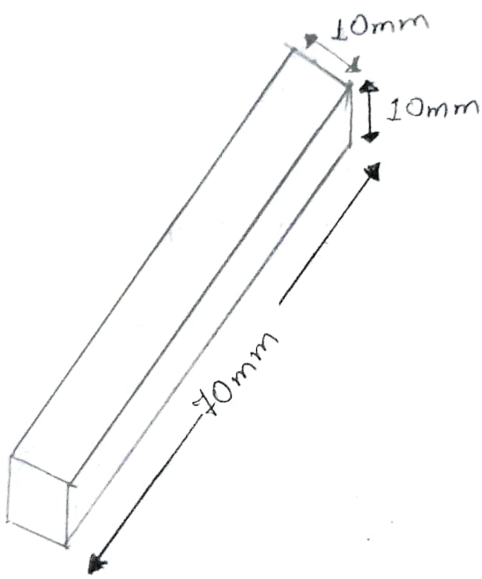
SMITHY - SHOP

Roll No. : AB-16030

TITLE : To prepare the job as per the drawing.

OBJECTIVES : To familiarise with the black-smithy work, tools and procedure.

DRAWING:



Raw material size :

All dia in m.m.

Length = 70 mm

Diameter: 10 mm

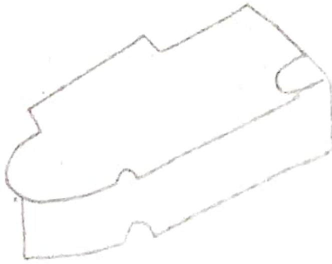
Tools	Particular	Specification
Hand Tools	Hammer	Cast iron, Cast steel
Cutting Tools	Anvil, Chisel	mild steel, Cast iron
Measuring Tools	Outside & inside puishing, scale etc.	Cast iron, steel blade.

Sequence of operations :

1.) Heat the job upto required temperature of 1300°C.
2.) One end is subjected to drawing out open on Anvil.
3.) Make the rod straight with hammer.
4.) Hammer is consequently struck on job to make it rectangular.
5.) Finishing of square of job is done by flat hammer.

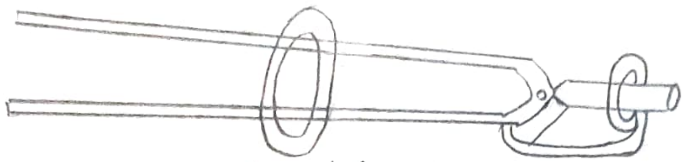
Precautions :

1.) All tools handling striking hammer must be used and handled with care.
2.) Cutting should be done by anvil pace.
3.) All operations must be finished in between 950°C - 1150°C.



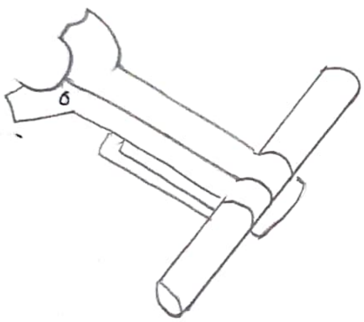
Sledge Hammer

Hand Hammer

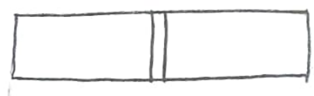


Clamping Ring

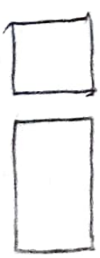
Hollow-Bit Tongs



Pick-Up Tongs



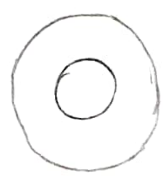
Cutting



Necking

Swagging

Bending



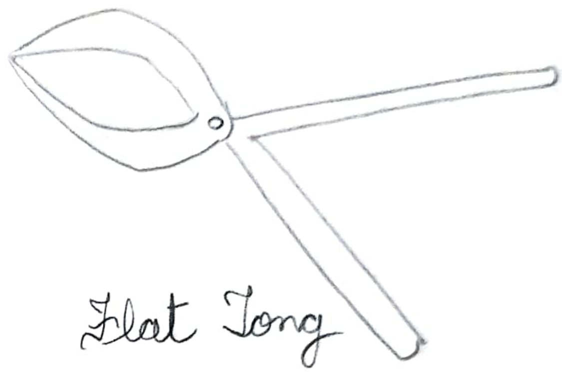
Forge Welding

Tapering

Hook

Drift

Wrench



Flat Tong

QUESTIONARIES :

1. Name the various kind of operation performed in Block-Smithy.
2. Indicate the statement given below
   TRUE or FALSE
   (a) Drawing operation is done by increase the length of stock at the Cost of its cross- section.
   (b) Up setting is used to increase cross-section.
   (c) Forged welding can be performed in a hot state.
   (d) Cutting angle of cold chisel is30 degree.
   (e) Bottom swages and fullers are inserted in the round hole of anvil. (Pritchethold).
3. Name the five type of joint which can be prepare fgrom forged welding.
4. What is the working temperature range of mild steel ?
5. What will be colours of Mild steel at hot working tempreture ?
6. Name the operation that you performed on a stock of 20 mm dia to make a bolt of 12mm dia.
7. What is difference between cold working and hot working.

Q1.) Name the various - - - - - - - Block-Smithy

⇒ (i) upsetting or Jumping (ii) Setting down

(iii) Riveting (iv) Bending

(v) Drawing down (vi) Cutting out

(vii) Swaging (viii) Welding

(ix) Fillering (x) Punching Drafting

Q2.) Indicate the - - - - - anvil. (Pritchethold)

⇒ a.) True b.) True c.) False

d.) False e.) false

Q3.) Name the - - - - - - - forged welding

⇒ (i) Butt joint (ii) Tap joint

(iii) U-joint (iv) Fork joint

(v) T-weld ~~or~~ jump weld

Q4.) What is the - - - - - - mild steel?

⇒ The temperature to begin working of mild steel is 1100°c - 1300°c and the temperature to finish forging is 800-875°c.

Q5.) What will be - - - - - working temperature?

⇒

Colour	Temperature (°c)
Faint red	500
Blood red	650

Cherry red	750
Bright red	850
Salmon	900
Orange	950
Yellow	1050
White	1200

Q6. Name the operation - - - - - - - of 12mm dia.

⇒ (i) Jumping (ii) Flattening

(iii) Drawing down (iv) Swage head to size

(v) Forge chamber on hand by a clapping tool, using the hush as support.

Q7.) What is difference - - - - - and hot working.

⇒

Cold Working	Hot Working
1.) It is done before the recrystallisation temperature.	1.) It is done after the recrystallisation temperature.
2.) Mechanical property decreases	2.) Mechanical property are improved.
3.) Internal & external residual stress develops in metal.	3.) Internal and external residual stress is not developed in metal.
4.) More amount of force is required.	4.) Lesser force is required.
5.) Deformation is limited.	5.) Deformation can take place.